

Test III (Classes 18-24)

Name
Date
Arithmetic

Instructions

- I) Write a title, name, date, and course.
- II) Number each problem, circle the number, rewrite the problem, and show all work.
- III) Put a rectangle around your answers.
- IV) Write any fractions in your answers in lowest terms and never leave your answers as improper fractions.

① Convert the following percents to decimals, then to fractions.

a) 7% b) 88% c) 0.00967% d) 7,056.1%

② Simplify the following operations.

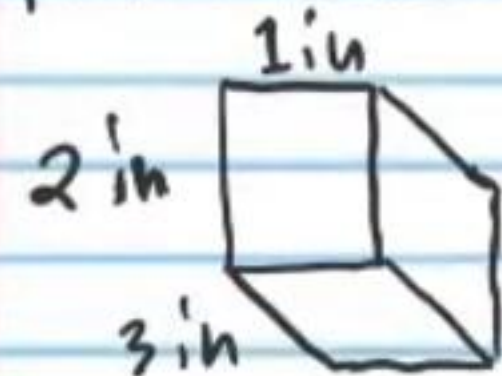
a) $4^2 - 3 \times (7 - 5) \div 2$ b) $[(4 - 1)^3 \div 9 + 1] \div 4 + 8$

c) $10 \div 2 \times 6 \times 3 \div 9$ d) $7 - 1 + 8 + 2 - 10$

③ The surface area of a rectangular prism is found with the following formula:

$$SA = 2 \times L \times W + 2 \times L \times H + 2 \times W \times H$$

Use this formula to find the surface area of the rectangular prism below.



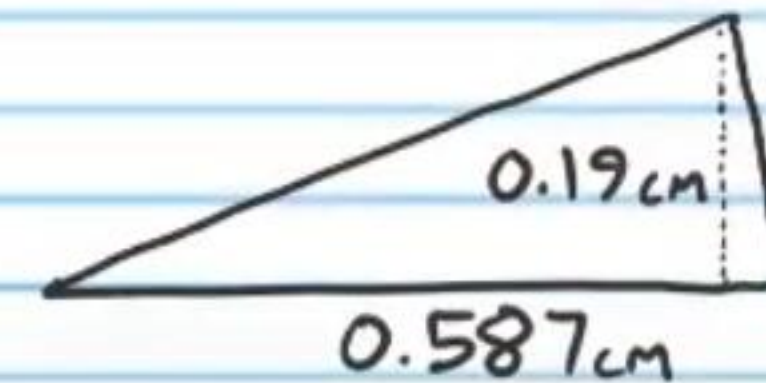
★ It doesn't matter which dimension is called length, width, etc., because you will get the same answer.

④ If there are 0.78 lbs of celery in a box that weighs a total of 1.2 lbs, what percent of the weight of the box is celery?

⑤ If 28 is increased by 59%, what is the result?

⑥ What is 19.1% of eight?

⑦ The area of a triangle is found with the following formula: $A = b \times h \div 2$. What is the area of the triangle below?



⑧ Alan's height grew by 3% during the summer. His new height is 57.68 inches. What was his height before he had this growth spurt?

⑨ Joshua collected a 0.36 ounce seashell, a 27.5 ounce rock, and a 1.5 ounce lobster shell. What is the total weight of the objects he collected?

⑩ Convert the following decimals to fractions, then to percents: a) 0.1 b) 3.7 c) 0.004503 d) 763.00009

⑪ You lend a friend \$280.20 at 5% simple interest paid monthly. How much will your friend owe you after six months?

⑫ 8% of a substance is platinum. If 4.8 kilograms of a substance is platinum, what is the total mass of the substance?

⑬ Convert the following fractions to decimals, then to percents.
a) $\frac{9}{10}$ b) $\frac{1}{100}$ c) $3\frac{5}{1000}$ d) $\frac{6680}{100,000,000}$

⑭ Carl bought a pair of shoes for \$55. The sales tax in his state is 6%. How much will Carl pay in tax? How much will he pay in total?

⑮ Jaime's pocket change decreased from \$0.8 to \$0.56. By what percent did Jaime's pocket change decrease?

Convert the following fractions to decimals, then to percents.

⑯ a) $\frac{1}{5}$ b) $\frac{7}{20}$

⑰ a) $\frac{1}{50}$ b) $\frac{11}{200}$

Convert the following fractions to decimals, then to percents.

⑱ $\frac{3}{8}$

⑲ $1\frac{3}{15}$

⑳ $\frac{4}{9}$

㉑ $\frac{1}{7}$

㉒ Frank's Furniture is giving a 15% discount on all items. If a customer buys a chair originally listed for \$84, how much will the customer pay before tax?

㉓ If the customer in problem #22 has to pay a sales tax of 10%, how much will he pay in total?

㉔ You take out an \$800 loan at 12% simple interest paid annually. If you keep the loan for twelve years,

how much will you owe in interest?

㉕ $\frac{7}{8}$ of a cake is flour. What fraction of the cake is not flour? What percent of the cake is flour? What percent is not flour?